

BHUJ, India

WMO: 426340

Lat: 23.288N

Lon: 69.670E

Elev: 268

StdP: 14.55

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	48.8	52.1	22.8	17.4	69.6	27.8	21.9	70.3	12.0	72.6	10.9	73.3	0.5	230	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	15.7	105.6	73.7	103.2	73.9	101.1	74.5	83.5	93.1	82.6	91.8	81.8	90.5	6.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.5	165.3	87.1	80.6	160.7	86.2	79.9	156.8	85.6	48.0	93.1	46.9	92.2	46.1	90.5	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
15.9	13.4	11.7	DB	47.5	110.0	5.1	2.6	43.9	111.9	40.9	113.4	38.1	114.9	34.4	116.8
			WB	40.1	85.3	3.6	0.9	37.5	85.9	35.3	86.4	33.3	86.9	30.7	87.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	81.9	68.0	73.4	82.2	87.9	90.5	90.2	86.3	83.9	85.1	86.1	78.6	70.4	(6)
(7)		DBStd	8.21	4.37	5.50	4.68	3.28	2.41	2.68	2.94	2.26	2.54	3.38	4.69	4.75	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	35	20	5	0	0	0	0	0	0	0	0	0	10	(9)
(10)		CDD50	11649	558	656	998	1136	1256	1206	1127	1052	1052	1118	857	633	(10)
(11)		CDD65	6209	113	242	533	686	791	756	662	587	602	653	407	177	(11)
(12)		CDH74	77221	1056	2386	6263	9188	11203	10853	8515	6770	7143	8238	4080	1526	(12)
(13)		CDH80	40610	250	1002	3575	5590	6831	6550	4114	2538	3240	4549	1924	447	(13)
(14)	Wind	WSAvg	4.1	2.2	2.7	3.3	4.9	7.1	6.7	6.8	5.7	4.1	2.2	1.7	1.9	(14)
(15)	Precipitation	PrecAvg	15.7	0.1	0.0	0.0	0.1	0.2	1.4	5.6	3.9	3.2	0.7	0.3	0.0	(15)
(16)		PrecMax	41.6	0.7	0.5	0.4	2.4	2.4	4.9	18.3	15.1	14.7	8.8	5.9	0.0	(16)
(17)		PrecMin	3.9	0.0	0.0	0.0	0.0	0.0	0.0	0.8	0.0	0.4	0.0	0.0	0.0	(17)
(18)		PrecStd	9.6	0.2	0.1	0.1	0.4	0.5	1.5	4.8	3.9	2.0	1.1	0.0	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.0	97.2	104.8	108.0	109.7	106.8	99.7	96.0	100.5	103.0	96.2	90.5	(19)
MCWB			66.1	68.6	69.8	72.0	75.1	78.0	80.6	79.6	76.4	72.5	68.7	68.4	(20)	
2%		DB	85.2	92.9	101.2	104.8	106.1	102.4	96.6	92.9	97.2	100.7	94.0	87.4	(21)	
		MCWB	64.0	67.3	69.7	71.0	75.4	79.9	79.8	78.2	77.3	72.4	69.0	66.4	(22)	
5%		DB	82.7	89.5	98.2	102.5	103.3	100.0	94.4	91.0	94.7	98.5	92.2	85.0	(23)	
		MCWB	63.4	66.2	69.2	71.2	75.5	80.0	79.5	78.2	77.5	72.4	68.7	65.1	(24)	
10%		DB	80.0	86.2	95.3	100.0	100.5	97.7	92.3	89.3	92.4	96.1	90.1	82.3	(25)	
		MCWB	62.0	64.4	68.9	71.3	76.5	80.0	79.3	78.1	77.4	71.8	67.4	63.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.8	74.7	81.0	82.2	84.0	84.6	83.9	83.5	83.9	81.9	78.1	73.6	(27)
MCD B			81.5	88.5	96.8	97.8	98.4	94.2	91.3	90.4	91.5	94.7	88.9	85.5	(28)	
2%		WB	69.1	71.6	76.3	79.3	82.5	83.5	82.7	81.8	82.0	79.8	75.5	70.7	(29)	
		MCD B	79.2	85.2	91.5	91.0	94.8	93.0	90.0	88.2	89.7	90.6	85.6	81.1	(30)	
5%		WB	66.6	69.5	74.2	77.7	81.5	82.6	81.8	80.7	80.7	78.2	73.4	68.6	(31)	
		MCD B	77.3	82.1	87.4	88.8	92.3	91.7	88.6	86.3	87.6	87.8	82.8	79.8	(32)	
10%		WB	64.2	67.7	72.3	76.3	80.6	81.8	81.0	79.8	79.6	77.1	71.1	66.3	(33)	
		MCD B	75.5	79.9	84.7	87.0	90.2	90.7	87.3	85.0	86.2	86.3	83.2	78.2	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	24.2	24.8	25.4	25.4	21.4	15.7	10.8	10.3	15.0	20.8	22.7	23.8	(35)
MCDBR			26.4	27.1	28.1	28.0	26.4	19.8	14.2	13.2	18.9	23.0	23.4	25.7	(36)	
MCWBR			12.0	10.6	8.9	7.5	7.0	4.1	2.9	3.1	4.4	7.0	8.9	11.6	(37)	
5% WB		MCDBR	24.6	24.6	26.0	24.7	20.9	16.0	11.8	10.7	14.2	18.9	20.9	23.8	(38)	
		MCWBR	14.0	11.6	11.7	7.8	5.8	4.3	4.0	4.3	5.5	6.9	9.8	13.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.412	0.434	0.495	0.541	0.598	0.707	0.759	0.685	0.589	0.536	0.517	0.444	(40)	
taud		2.123	2.080	1.926	1.854	1.780	1.609	1.548	1.679	1.853	1.882	1.882	2.031	(41)		
Ebn at Noon		266	269	258	248	233	207	196	211	230	235	229	249	(42)		
Edh at Noon		45	50	61	67	71	84	89	78	65	60	57	48	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1407	1676	1965	2179	2221	1891	1387	1387	1689	1724	1431	1319	(44)	
(45)		RadStd	48	49	67	57	65	130	109	93	114	72	66	67	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	+0.88	+4.05	+4.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+0.41	N/A	+3.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page